

Exploratory Data Analysis

Principal Component
Analysis (PCA)

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Today's Agenda

Concept, intuition, output, and hands-on analysis

01

Why EDA?

Data avalanche
Not modelling
Geometric intuition

02

PCA: The Idea

Finding new axes
Standardisation
Variance explained

03

PCA Output

Factor maps
Correlation circle
Biplots · Supplements

04

PCA in R

FactoMineR
factoextra
Full workflow

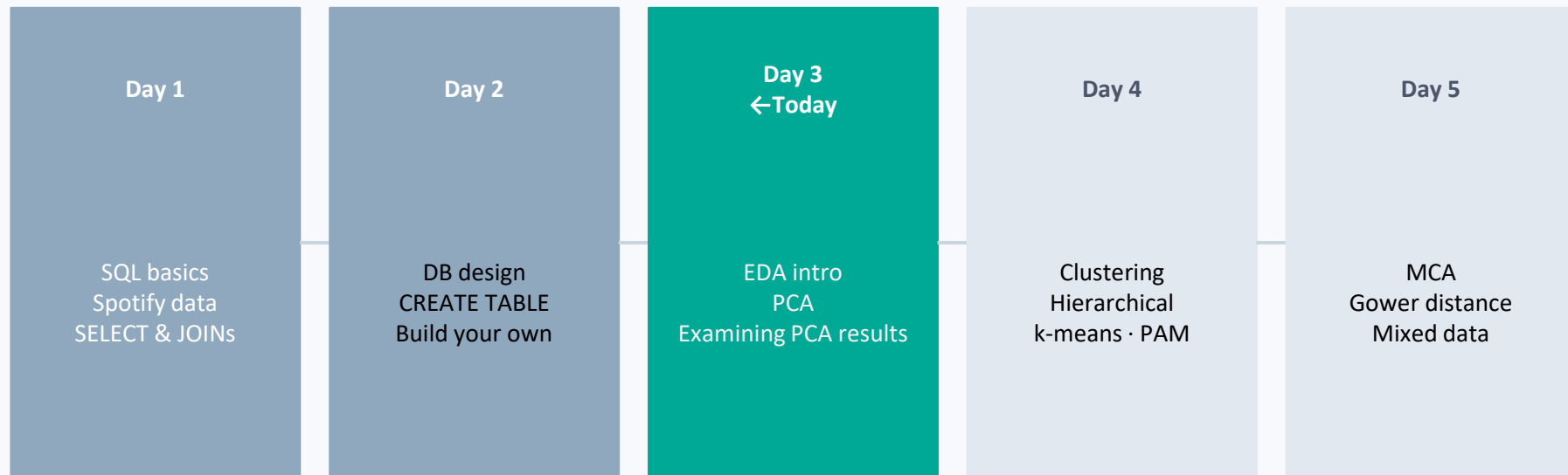
05

Lab

Spotify genres
Spotify tracks
Valkalkylator

Where We Are in This Course

SQL is the foundation — now we analyse what it gives us



The SQL queries from Days 1–2 produce the data matrices we analyse today. The pipeline is: database → SQL query → data frame → PCA.

1

Why exploratory data analysis?

What are we trying to do?

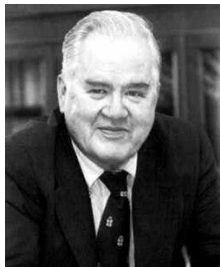


The Data Avalanche: Why We Need These Methods

You have 89,740 Spotify tracks, each described by 9 audio features. You have 1,600 political survey respondents, each with 28 policy positions. You have a survey where respondents ask 20 different questions. Et cetera!
You need methods that reveal the structure!

John Tukey, 1977

Exploratory Data Analysis is detective work: finding clues, building hypotheses, looking at data in many ways before you decide what you are looking for.



“It is important to understand what you CAN DO before you learn to measure how WELL you seem to have DONE it”

Explore first! Confirm later. Most datasets hold surprises.

If you skip exploration, going straight to measurement, you risk measuring wrong things very precisely!

What we will cover in Days 3–5: Tools for EDA!

PCA

Dimensionality reduction for continuous standardized variables

Clustering

Grouping observations by similarity

MCA + Gower

Extending the toolkit to categorical and mixed data

How is EDA different from Modelling?

Different question, different standards, different risks

	Exploratory (EDA)	Confirmatory (modelling)
Goal	Discover unexpected structure	Estimate parameters, test hypotheses
Ground truth	None (there is no correct answer to find!)	Yes — the true parameter exists
Output	A visualisation, a typology, a hypothesis	Coefficients, p-values, confidence intervals
Validation	Interpretability and substantive coherence	Statistical significance, model fit
Risk	Over-fitting structure that is not really there	Type I/II errors, incorrect causal claims
Your job	Describe and interpret what you find, honestly	Report uncertainty in your estimates

EDA results are hypotheses, not conclusions. The appropriate response to an interesting cluster is: 'this suggests we should investigate further', not 'this proves'.

Geometric Data Analysis

All three days share this underlying shared intuition

Each observation is a point in a high-dimensional space: one dimension per variable. EDA methods find the best low-dimensional view of that space, preserving as much structure as possible.



PCA

PCA

Day 3 (today)

Continuous variables. Rotates the original axes to find the directions of maximum variance. Gives a 2D map of observations and variables.



K

Clustering

Day 4

Groups observations into clusters based on similarity. Builds on the PCA space or on a distance matrix computed directly.



MCA

MCA + Gower

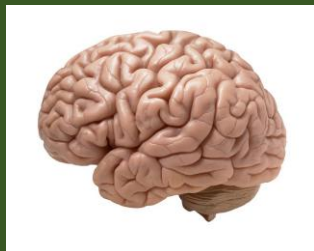
Day 5

Extends the geometric approach to categorical and mixed-type data. Same intuition, different underlying math.

2

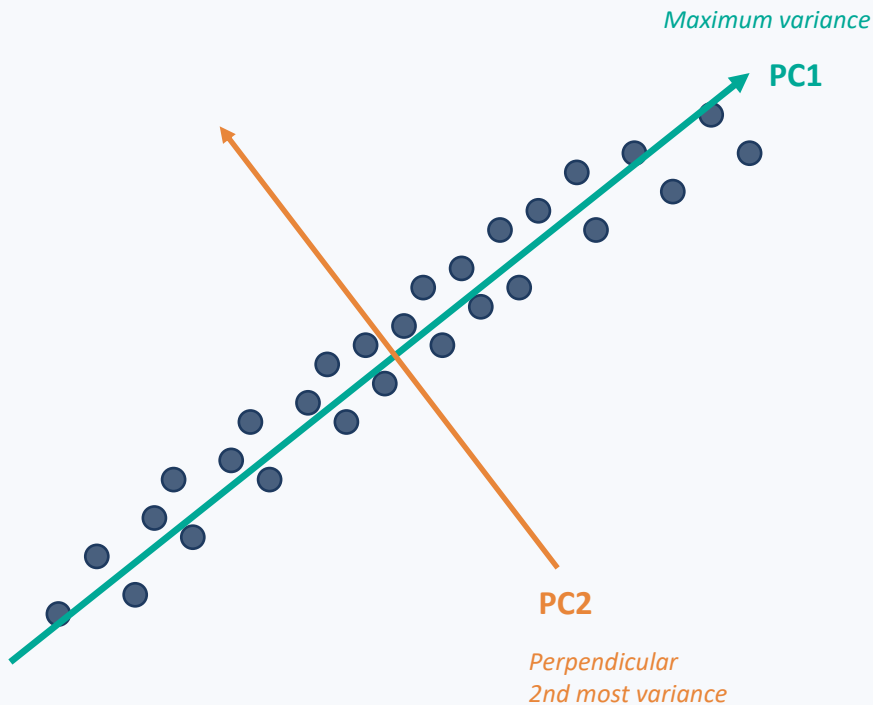
PCA: The idea!

Finding new axes · Standardisation · Variance explained



PCA in Two Dimensions — The Intuition

Finding the axis of maximum variance in a point cloud



Start with the data cloud

Each observation is a point in variable space. With 2 variables you have 2D. With 9 variables: 9D (!)

PC1 = direction of max variance

A new axis through the cloud in the direction that captures the most spread. This is a rotation of the original axes. Each PCA: a blend of original variables!

PC2 = perpendicular to PC1

Captures the most remaining variance, subject to being at 90° to PC1. The components are uncorrelated by construction.

Each PC is a weighted sum

$PC1 = w_1 \times Var1 + w_2 \times Var2 + \dots$ These weights are the loadings: they tell you what each component means.

From 2D to n Dimensions (and why standardisation is mandatory)

The key assumption that PCA makes

Extending to 9 dimensions

With 9 audio features, each track is a point in 9-dimensional space.

PCA finds new axes in exactly the same way:

- PC1: direction of maximum variance across all 9 dimensions
- PC2: next, subject to $\perp$ to PC1
- ... up to PC9 (= original dimensionality)

We keep only the first 2–3. They summarise most of the structure — which is only possible if the data has low intrinsic dimensionality. (Sometimes this doesn't work!)

If all 9 variables were random and uncorrelated, PCA would not help — each axis would explain 1/9 of the variance and there would be nothing to summarise.

Why standardisation is mandatory

PCA finds directions of maximum variance. If variables are on different scales, the largest-scale variable dominates — not because it is more important, but because it has bigger numbers.

Spotify data without standardisation:

```
tempo:      range 0-220 BPM → large variance
loudness:   range -60 to 0 dB → large
danceability: range 0-1 → tiny
valence:    range 0-1 → tiny
```

```
→ PC1 will be almost entirely tempo + loudness.
   Not because they matter more — because
   they have bigger numbers.
```

Solution: subtract the mean, divide by the standard deviation (scale.unit = TRUE in FactoMineR). Every variable now has mean 0 and variance 1.

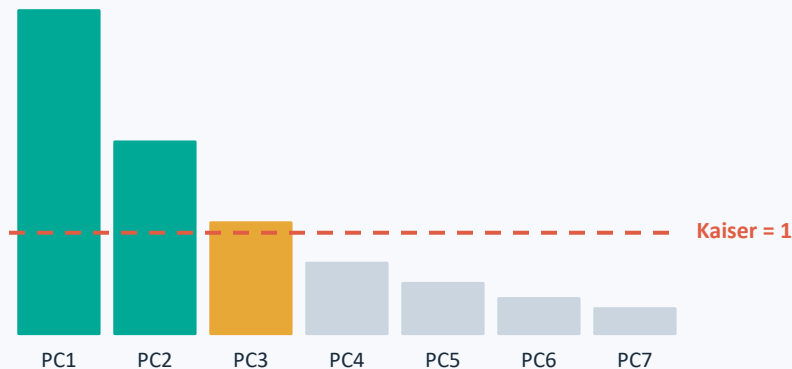
Measuring Variance Explained: Eigenvalues and the Scree Plot

How much of the original information does each component capture?

What eigenvalues represent:

After standardisation, each variable contributes exactly 1 unit of variance. The total variance is equal to the number of variables (9 in our example)

PCA redistributes this variance across the new components. The eigenvalue of PC1 is the variance it captures. If PC1 has eigenvalue 3.2, it captures $3.2/9 = 35.6\%$ of the total variance.



Three ways to choose how many PCs

Elbow rule

Look for the 'elbow' in the scree plot — where the curve flattens. Keep the components before the drop-off.

Kaiser rule

Keep components with eigenvalue > 1 (the dashed line). Logic: each component should explain more than a single original variable. Common but unreliable — tends to keep too many components.

Interpretability

Keep the number of components you can give a meaningful name to. This is often 2–3. If you cannot interpret PC4, retaining it adds noise.

Parallel analysis (psych::fa.parallel) is more rigorous: compares eigenvalues to those from random data of the same size. Worth knowing about.

Why do we do PCA and similar techniques?

Ok, this is all cute: but why do we do this?

Three reasons for the bigger picture of analysis

1. Exploration and description

This is what we will do in the lab. You use PCA to understand the structure of your data: which variables hang together, what the main axes of variation are, where observations sit relative to each other. The output is insight, not further analysis: PCA as an end in itself!

2. Dimensionality reduction for subsequent analysis

You replace your original n variables with a smaller number of component scores, then use those scores as input to another method. For instance: a survey on political opinions. Instead of working with 28 variables, you work with 2-3 component scores that capture most of the variance. This is PCA as a preprocessing step.

3. Dealing with multicollinearity

In regression, highly correlated predictors cause problems. PCA components are by construction uncorrelated with each other, so replacing correlated variables with components before regression sidesteps the problem.



```
# Extract individual scores (one row per respondent,  
# one col per component  
scores <- as.data.frame(pca_res$ind$coord) |>  
rownames_to_column("respondent_id")  
  
# Now each respondent has a position on PC1 and PC2!
```

3

Reading PCA output

Individual maps · Correlation circle · Biplots · Supplements

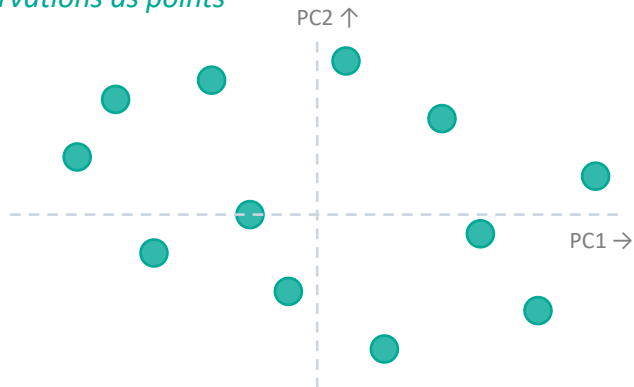


PCA Produces Two Complementary Outputs

Observations and variables — both get coordinates

Individual Factor Map

Observations as points



Each genre / track / respondent is a point.

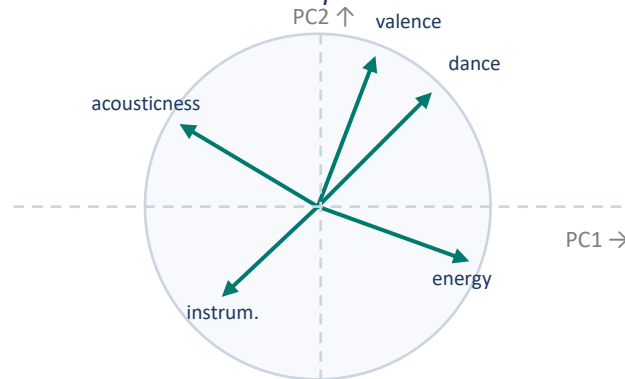
Proximity = similarity in the original variable space.

Points that are close share similar audio profiles (or similar policy positions).

Points that are far apart are very different.

Variable Factor Map

Variables as vectors in the same space



Each original variable is a vector.

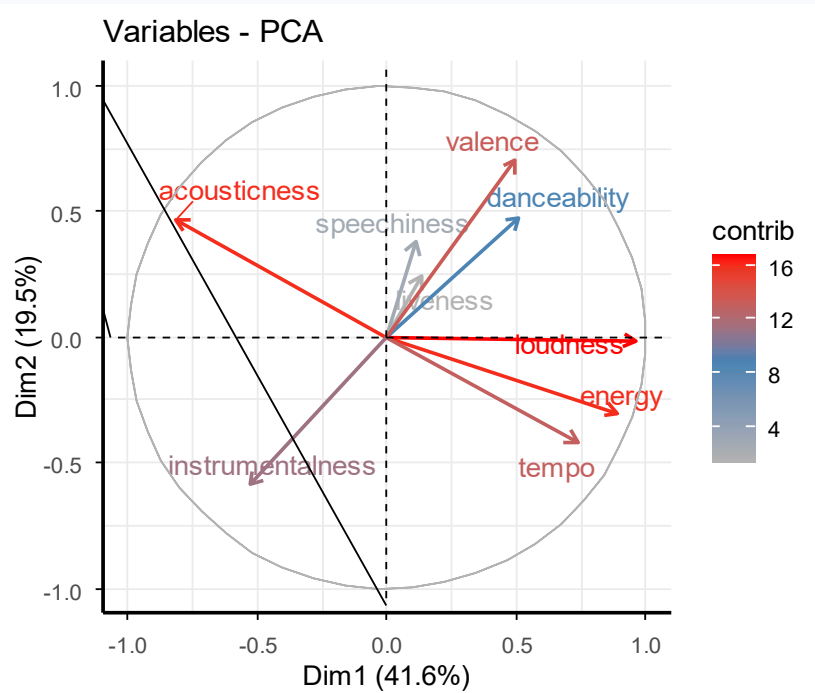
Length = how well represented in this 2D view.

Direction = which components it loads on.

Angle between vectors = correlation between variables.

The Variable Factor Map (Correlation Circle)

Reading the relationships between variables



How to read the correlation circle:

Length of the vector

Long vectors (reaching the circle) are well-represented in this 2D view. Short vectors may be explained by other components.

Vectors pointing together

energy and loudness → same direction = positively correlated.
Tracks with high energy tend to be loud (and also have high tempo)

Vectors pointing opposite

energy vs acousticness → opposite directions = negatively correlated.
Acoustic tracks tend to be low energy.

Vectors at 90°

Roughly uncorrelated. Knowing a track's danceability tells you little about its acousticness.

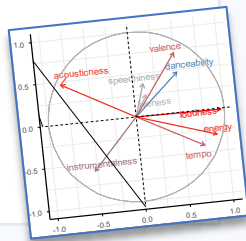
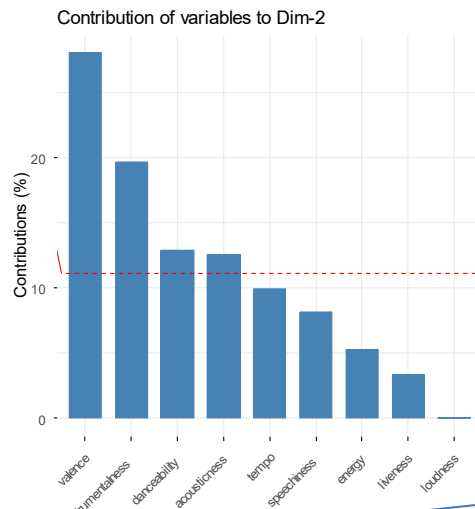
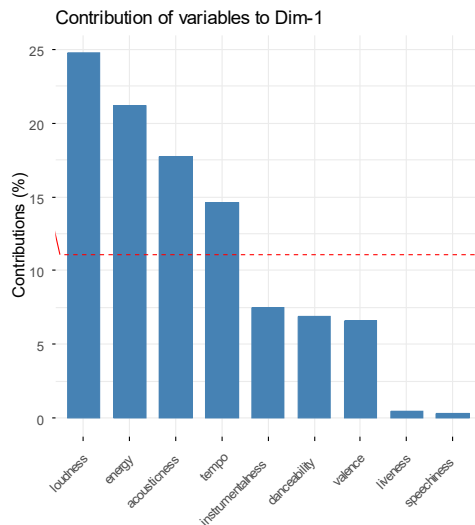
Reading the axes

Variables pointing right load on PC1 (+), left on PC1 (-). Up on PC2 (+), down on PC2 (-). Use this to name the components!

Contribution Plots — What Drives Each Component?

Naming the axes is the core interpretive task

PC1 & PC2: Contribution of each variable (also see Variable Factor Map)



From contributions to axis names

PC1: high contribution from loudness, energy, acousticness and tempo. Might point in opposite directions on the correlation circle (e.g. acousticness)

→ Name: "Acoustic–Electric" or "Quiet–Intense" dimension. "Amped–Quiet"?

PC2: high contribution from valence, then instrumentalness pointing opposite direction

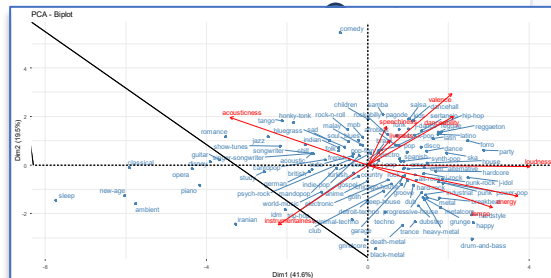
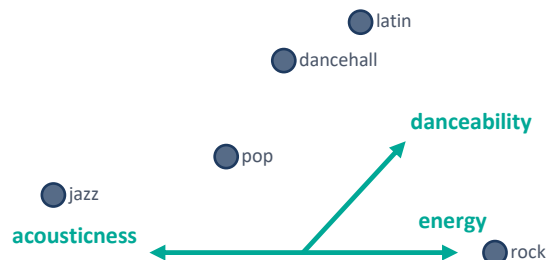
→ Name: "Mood and Groove" or "Danceability–Positivity" dimension

This naming step is a social science act — the method provides coordinates, you provide meaning. Two researchers can legitimately name the same axis differently. CHECK the Variable Factor Map!

The Biplot and Supplementary Variables

Combining both outputs and adding external information

The biplot: Individuals + variables together



Genres near the tip of the 'acousticness' arrow → highly acoustic.
Genres opposite it (metal, rock) → high energy/loudness.

Supplementary variables: Project without influencing

quali.sup — categorical

A genre label, party affiliation, country region. Projected as a point at the centroid of all individuals with that category. Shows where each category sits in the PCA space without distorting the axes.

quanti.sup — continuous

An additional numeric variable that was not used to build the axes. Projected as a vector. Shows how it correlates with the PCA dimensions.

ind.sup — individuals

Observations not used to compute the axes. Projected onto the space afterwards. Useful for: respondents with many missing values, new data points, or a test set.

In R: `PCA(data, quali.sup = c(1), quanti.sup = c(2,3), ind.sup = c(51:60))`

Break!



4

PCA in R with FactoMineR

PCA() · fviz_eig · fviz_pca_var · fviz_pca_biplot



The Full PCA Workflow in R

From SQL query to PCA in one pipeline (here you are given SQL commands)

```
# Libraries needed: DBI, RSQLite, FactoMineR, factoextra

# 1. Connect with database, pull genre profiles from the database (SQL from Day 1)
con <- dbConnect(RSQLite(), "data/music.sqlite")

# 2. Query database, get 'genre_profiles' data frame back
genre_profiles <- dbGetQuery(con, "
  SELECT g.name AS genre,
         AVG(t.danceability) AS danceability, AVG(t.energy) AS energy,
         AVG(t.loudness) AS loudness, AVG(t.speechiness) AS speechiness,
         AVG(t.acousticness) AS acousticness, AVG(t.instrumentalness) AS instrumentalness,
         AVG(t.liveness) AS liveness, AVG(t.valence) AS valence,
         AVG(t.tempo) AS tempo
  FROM tracks t
  JOIN track_genres tg ON t.track_id = tg.track_id
  JOIN genres g ON tg.genre_id = g.genre_id
  GROUP BY g.name")

# ...and close the connection
dbDisconnect(con)
```

See lab_pca_music_genres.R in today's Lab folder on Lisam!

The Full PCA Workflow in R

From SQL query to biplot in one pipeline

```
# 2. Run PCA (scale.unit=TRUE standardizes variables automatically)
pca_res <- PCA(
  genre_profiles,
  scale.unit = TRUE,
  quali.sup = 1,      # 'genre' column is a label, not a variable (but could be made row header instead; see lab)
  graph      = FALSE
)

# 3. Inspect
summary(pca_res)
```

graph = FALSE suppresses the automatic plots so you can control them with `fviz_*` functions.

quali.sup = 1 tells PCA that column 1 is the genre label; it gets projected onto the space but does not influence the axis computation. (But in lab script, we transform this into row name instead.)

Visualising PCA Output with factoextra

Five functions you will use in every PCA

```
# Scree plot – variance explained per component
```

```
fviz_eig(pca_res, addlabels = TRUE, ncp = 9)
```

```
# Variable factor map (correlation circle)
```

```
fviz_pca_var(
```

```
  pca_res,
```

```
  col.var = "contrib",      # colour by contribution
```

```
  gradient.cols = c("grey70", "steelblue", "red"),
```

```
  repel = TRUE              # avoid label overlap
```

```
)
```

```
# Individual factor map (genre points, labelled)
```

```
fviz_pca_ind(pca_res,
```

```
  label = "all",
```

```
  repel = TRUE
```

```
)
```

```
# Contribution plot: which variables drive PC1?
```

```
fviz_contrib(pca_res, choice = "var", axes = 1)
```

```
# axes=2 for PC2
```

```
# Biplot – observations and variables together
```

```
fviz_pca_biplot(pca_res,
```

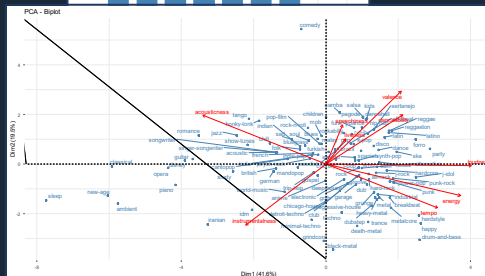
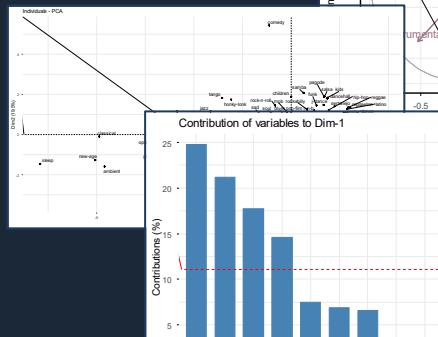
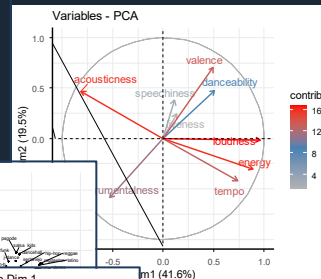
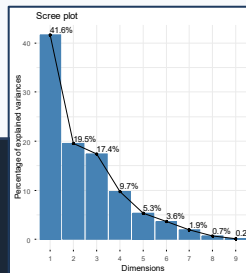
```
  label = "var",
```

```
  col.ind = "steelblue",
```

```
  col.var = "red",
```

```
  repel = TRUE
```

```
)
```



From PCA to Clustering (the next step)

Looking at the individual factor map, you will notice that some genres cluster together visually: perhaps classical and ambient on one side, metal and punk on another. Day 4 formalises this using hierarchical clustering and k-means. But let's check the in-built cluster function in FactoMineR:

Preview: HCPC (Hierarchical Clustering on Principal Components)

```
# FactoMineR has a dedicated function that clusters directly in PCA space
# It runs hierarchical clustering on the PC scores and then
# consolidates with k-means – the best of both approaches
hcpc_res <- HCPC(pca_res,
  nb.clust = -1, # -1 = choose automatically from dendrogram
  graph    = FALSE)

# Plot: individuals coloured by cluster
fviz_cluster(hcpc_res,
  repel      = TRUE,
  show.clust.cent = TRUE,
  main = "Genres – clusters in PCA space")

# Which genres are in each cluster?
hcpc_res$data.clust |>
  select(genre, clust) |>
  arrange(clust)
```

5

Lab 3

Spotify genres · Spotify tracks · Valkalkylatorn



Lab 3 — PCA on Three Datasets

Work individually or in pairs · ~30 min per dataset

Part A

Spotify Genre Profiles

- SQL query → genre × feature matrix
- Run PCA, produce scree plot
- Variable factor map: which features cluster?
- Individual map: name PC1 and PC2
- Biplot combining both

Lab file:

lab_pca_music_genres.R

(work through: all there!)

Part B

Spotify Track Sample

- SQL random sample of 5,000 tracks
- Exclude key, mode, time_signature
- Same PCA workflow: now 89k points
- How does the result compare to genre-level?
- What does an unlabelled point cloud add?

Lab file: NONE!

(but use the slides and the one for genres)

Part C

Valkalkylator Survey Data

- SQL query from valkalkylator.sqlite
- Data: how 9 political parties agree/disagree on 28 statements
- PCA on these 28 agreement scores
- What drives PC1? Can you name the axis?

Lab file (skeleton):

lab_valkalk_party_responses.R

(supplement with what you learned here)

Lab deliverable: for each dataset, write 2–3 sentences naming and interpreting PC1 and PC2 in substantive terms.

Lab Part B: Obtaining Spotify Tracks Sample

5,000 individual tracks: unlabelled points in PCA space

```
# — B1. Random sample from the database —————
con <- dbConnect(SQLite(), "data/music.sqlite")
tracks_sample <- dbGetQuery(con, "
  SELECT track_id,
         danceability, energy, loudness, speechiness,
         acousticness, instrumentalness, liveness, valence, tempo
  FROM tracks
  ORDER BY RANDOM()      -- random sample without loading everything
  LIMIT 5000")
dbDisconnect(con)

# — B2. Run PCA —————
# Exclude track_id — it's an identifier, not a variable
pca_tracks <- PCA(tracks_sample[, -1], scale.unit = TRUE, graph = FALSE)

# — B3. Explore —————
fviz_eig(pca_tracks, addlabels = TRUE)
fviz_pca_var(pca_tracks, repel = TRUE, col.var = "contrib")
fviz_pca_ind(pca_tracks, geom.ind = "point", # no labels — too many points
             alpha.ind = 0.3,               # transparency helps
             col.ind   = "steelblue")
```

Variables to exclude

key (0–11, categorical by nature), mode (binary: major/minor), time_signature (discrete, usually 4). These are not interval-scale continuous variables.

What changes vs Part A

89k unlabelled points instead of 114 named genres. The variable map should be similar. The individual map looks very different — a dense cloud. You must read the variable map to interpret the space.

Valkalkylatorn 2018 Party Affiliation Test

www.valkalkylatorn.se

Hobby project, then developed for
Expressen newspaper magazine

This iteration: collect data

Novelty: opinion, AND significance

~1500 respondents, 28 statements

Also available for 2nd assignment!

Please do not share data (unpublished)!



VALKALKYLATOR 2018

1/28

Klimatpolitiken ska överordnas annan politik

Vad tycker du?

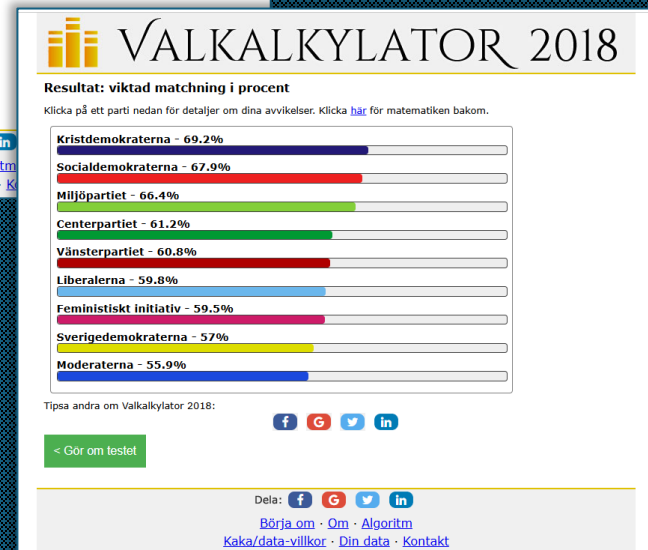
Helt emot Helt för

Hur viktigt tycker du detta är?

Helt oviktigt Mycket viktigt

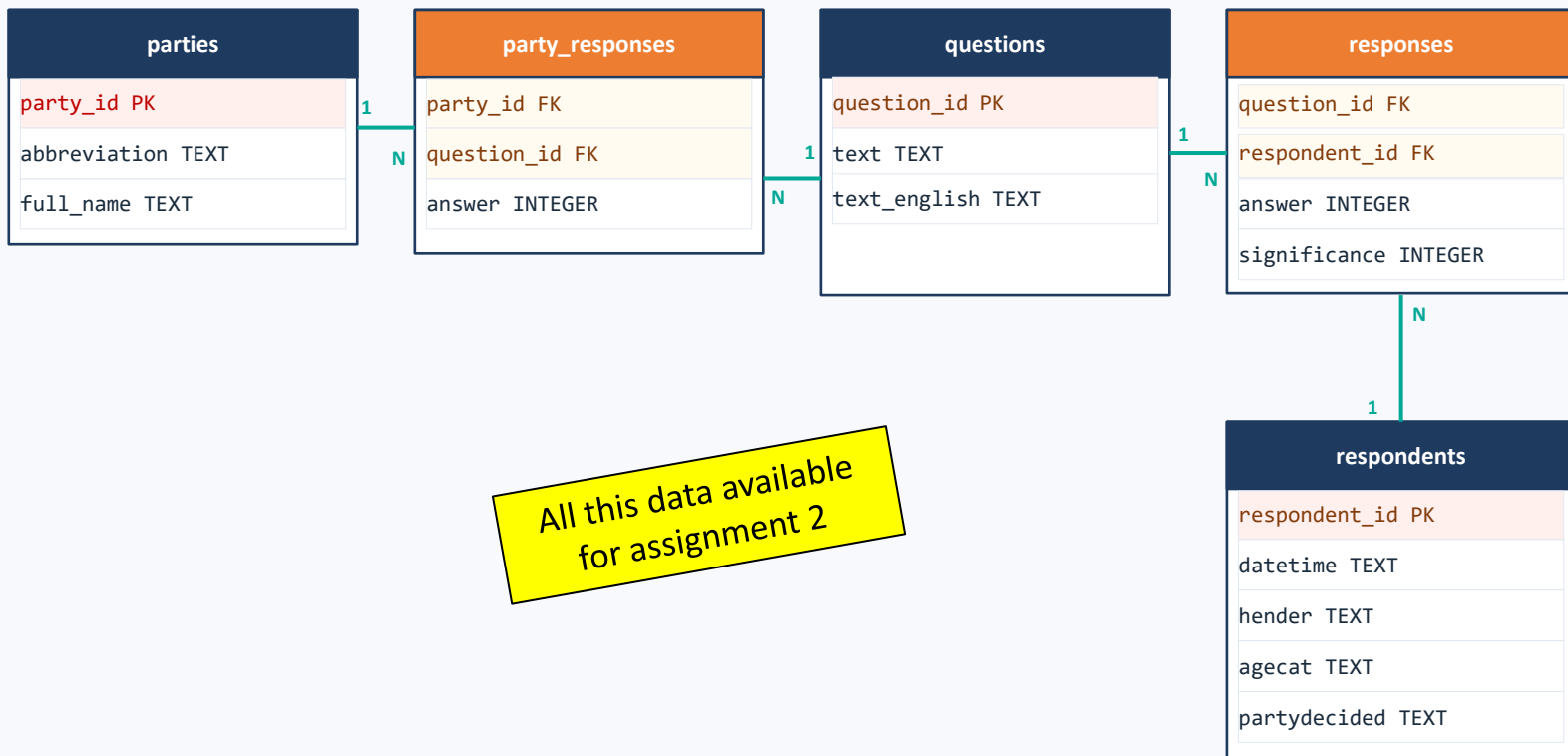
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[Kaka/data-villkor](#) · [Din data](#) · [Kontakt](#)



Valkalkylatorn: Entity-Relationships diagram

Three entities – parties, questions, respondents – and two junction tables binding respondents to questions, and parties to questions



All this data available
for assignment 2

PK Primary key

FK Foreign key

Lab Part C — Valkalkylator Survey Data

1,600 respondents × 28 policy positions — your own collected data

The valkalkylator.sqlite database uses the same structure you learned to query in Days 1–2. You are provided with a SQL query to obtain a dataframe from this database

```
# Valkalkylator exploration# You will examine the party-as-respondent matrix
# 9 political parties responded to 28 statements, from agree to disagree
# on a 0-4 likert scale (rescaled to 0-1)

# We will get the data from /data/valkalkylator.sqlite

# See:
# lab_valkalk_party_responses.R
```

Supplementary variables: project age_group and gender onto the PCA space using quali.sup. Do older respondents sit in a different region?

What to Look For — Interpretation Guide

The technical output is the starting point, not the endpoint

1

Scree plot

How many components explain ~70–80% of variance? Where is the elbow? Does the Kaiser threshold agree? Choose your number.

2

Variable factor map

Which variables cluster together (similar direction)? Which oppose each other? Which are long vectors (well-represented)?

3

Name the axes

Look at which variables load positively and negatively on PC1. Give PC1 a substantive name. Repeat for PC2. Write one sentence each.

4

Individual map

Do groups appear? Are there outliers? Do known groupings (genre families, political parties) correspond to regions of the map?

5

PCA for Valkalk?

The original party answers were coded as 5-step likert between 0 and 4, then recoded to 0, 0.25, 0.5, 0.75 and 1. MCA is then technically more suitable.

EDA results are descriptions, not conclusions. The appropriate output is: "PC1 appears to represent X, with genres/respondents that score high on PC1 tending to be Y. This suggests a further investigation of Z."